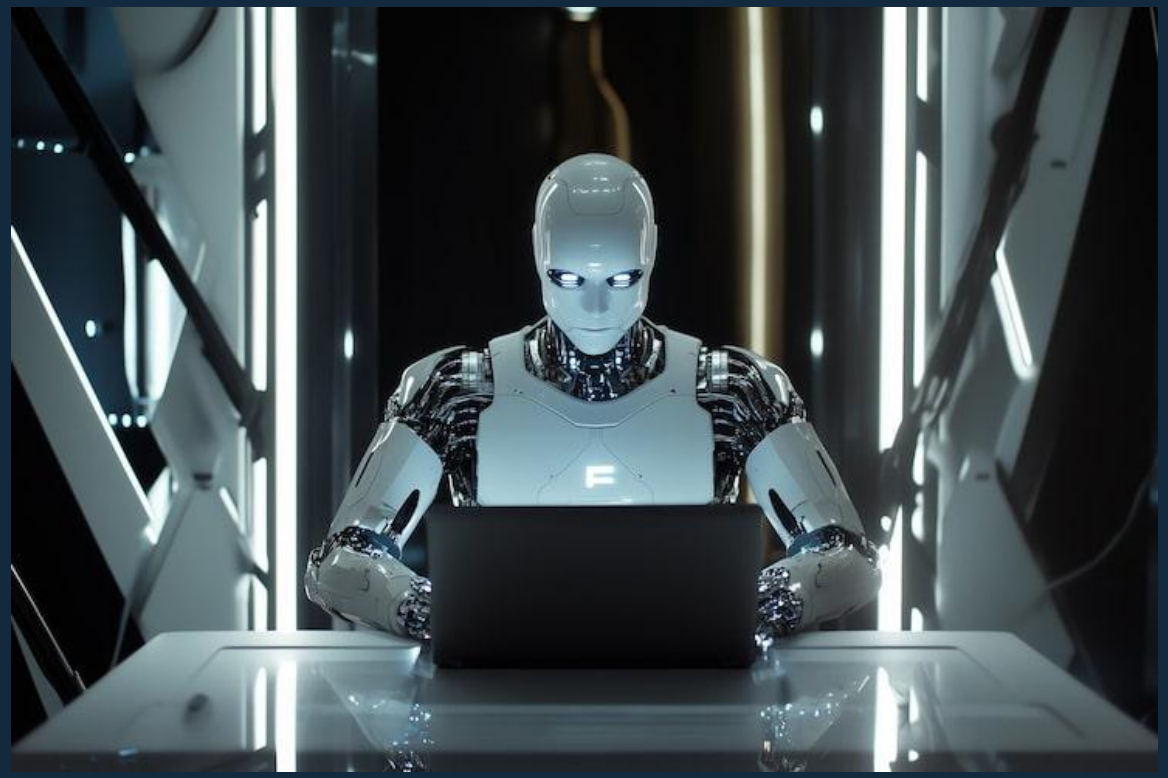
AI-Enhanced Web-Based Computer Laboratory Management System for

Palawan Technology College Inc.



Project Overview

A web-based system that puts laboratory management in one place.

It replaces paper logbooks, spreadsheets, and verbal coordination in PTCI's computer laboratories.

It is called AI-enhanced because the prototype shows how usage insights, maintenance alerts, and a help assistant could support staff in planning and decision making.

FOR
Laboratory staff · Instructors · Students · Administrators



Lab schedules



Computer unit status



Equipment records



Problem reports

Problem Statement



Schedule clashes

How can scheduling conflicts and unrecorded reservations be minimized?



Unclear unit status

How can computer unit status be monitored more effectively?



Untracked problems

How can laboratory issues be reported and tracked efficiently?



No usage overview

How can administrators access accurate information for decision-making?



The need: one place to reserve labs, check availability, report issues, and view lab information, so the school can anticipate problems instead of only reacting to them.

Project Objectives



GENERAL OBJECTIVE

Design a user-friendly prototype of an AI-enhanced Computer Laboratory Management System that shows how PTCI can manage its labs more efficiently.

SPECIFIC OBJECTIVES

1. Identify the needs and requirements of the users and develop a web-based scheduling system to prevent conflicts and improve laboratory utilization.
2. Design a user-friendly interface for laboratory scheduling, computer monitoring, and issue reporting.
3. Develop a web-based prototype of the proposed Computer Laboratory Management System.
4. Integrate AI-powered features such as usage insights, maintenance alerts, and an AI Help Assistant to enhance laboratory management.

Who Will Use It



Laboratory staff / custodian

Manage schedules, check computer status, update equipment records, and handle problem reports



Instructors

Check lab availability, reserve a lab for a class, and report equipment problems



Students

View lab schedules and open hours, check availability, and report a faulty unit



Administrators / IT office

View usage summaries, maintenance alerts, and reports to support decisions

Proposed Features



Laboratory dashboard

Today’s schedules, available labs, computers in use, and open problem reports



Schedule and reservation

Calendar view with a reservation form that warns when a time slot is taken



Computer and equipment status

List or floor plan of each lab’s computers: available, in use, or defective



Problem reporting

Report a faulty unit and track its status: reported, in progress, fixed



AI usage insights

Peak lab hours and a maintenance alert for units with repeated problems



AI help assistant

Chat-style screen answering questions such as which lab is free on a given day

Scope and Limitations

In scope

Pages and interactions the prototype will demonstrate

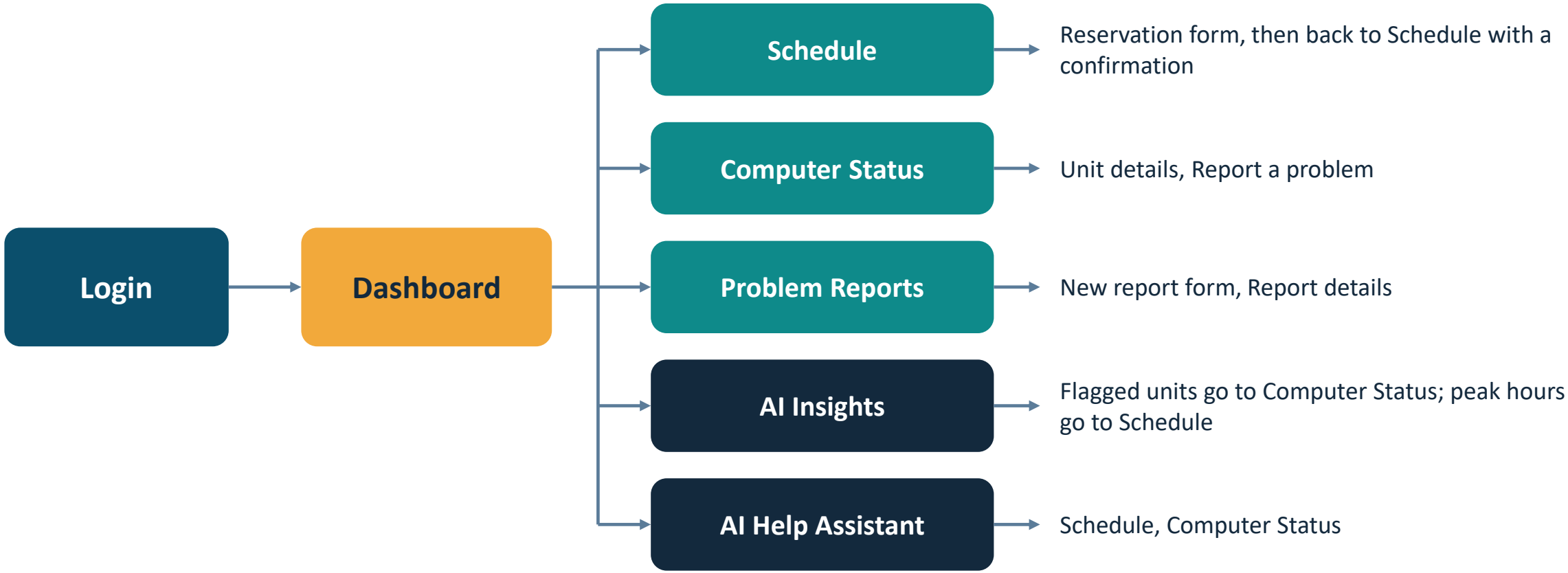
- **Login screen**
- **Dashboard**
- **Schedule and reservation**
- **Computer status view**
- **Problem report form and list**
- **AI insights screen**
- **AI help assistant screen**
- **Navigation paths per user type**

Limitations

What the prototype will not include

- Prototype only: no database integration, actual transactions, authentication, or live data processing, unless the instructor approves
- AI features appear only as designed screens with sample content
- No machine learning model is trained or connected

Navigation Flow



The flow starts at login and branches by user type.

Prototype Design and Tools

Flowchart

Shows how users move between screens, starting at login and branching by user type.

Figma or draw.io

Low-fidelity wireframes

Basic layout sketches of each key screen: header, side menu, content areas, tables, buttons.

Figma

Clickable prototype

Designed screens with school colors and logo, linked so navigation can be demonstrated.

Figma, HTML, CSS, JavaScript, Bootstrap

STATUS COLORS

Available **In use** **Defective**

ALSO USED

Canva: posters, logos, presentation graphics

Sample data: spreadsheet of made-up schedules and reports

Final tools follow instructor approval

Expected Output

A presentable software application prototype that demonstrates the system's:



Interface



Navigation



Major features



User workflow

The prototype does not need a live database or a fully working application.